

Consumer Decision-Making Analysis in R

Reproducible, Collaborative, and AI-Aware Workflows

Overview

Structure of the workshop

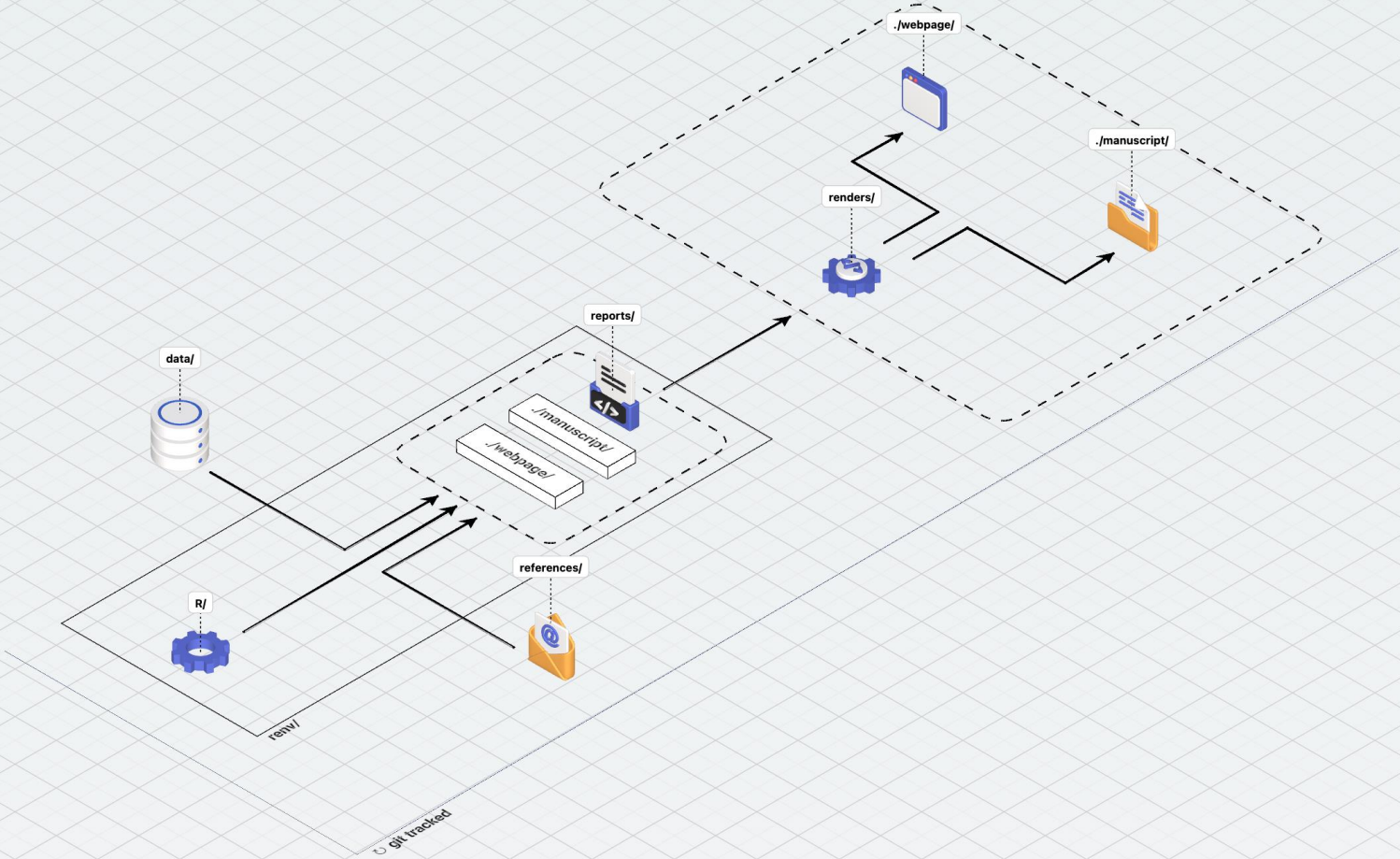
- ☒ Day 1: The scaffold
- ☐ Day 2: R and data analysis
- ☐ Day 3: Producing a report + Supercharge

Before you leave today

- [✓] GitHub account created
- [✓] Repository forked to your account
- [✓] GitHub Pages on → your website loads
- [✓] First commit made (README link points to your site)
- [✓] R and RStudio installed
- [✓] Fork cloned and opened via the .Rproj
- [✓] `renv::restore()` finished successfully
- [✓] `quarto render reports/webpage` works locally



Achievement unlocked
Software Developer Lvl 1



The three-day roadmap

Day 1: Scaffold

- R
- Rstudio
- Git/GitHub basics
- Folders
- Data protection
- Quarto

Day 2: Analyse

- Base R
- Import
- Clean
- Transform
- Visualise
- Describe
- Model
- Moderate / mediate
- Collaborate

Day 3: Publish

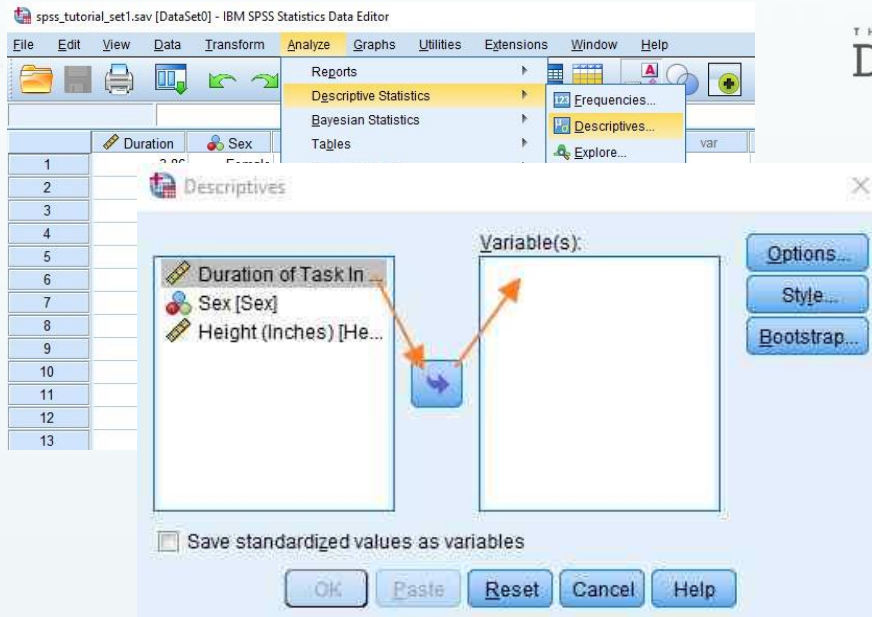
- Export tables and figures
- Write methods
- Write results
- Static/ Annotated/ interactive report
- AI-use statement
- Reproducibility check

Day 2

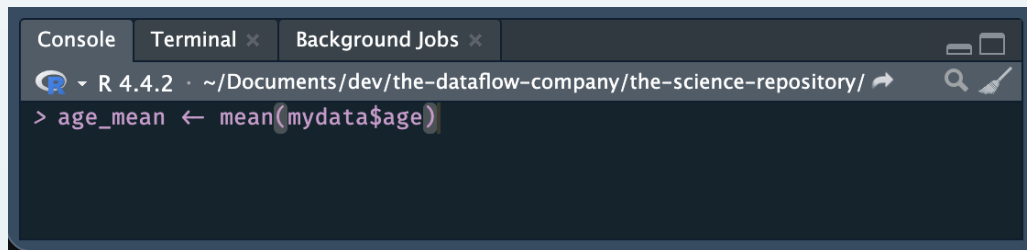
R & Data Analysis

The How

Buttons (Graphical User Interface)



Console (Command Line Interface)





user

Interfaces

Console (CLI)

`mean(mydata$age)`

Button (GUI)

Clicks: Analyze > Descriptive Statistics >
Descriptives > add variable(s) > ok

same

Instruction (expression)

`mean(mydata$age)`

R Engine / interpreter

1. lookup ``mydata`` in the workspace
2. lookup ``$age`` in ``mydata``
3. calculate mean of ``$age``

Output

`#> 41`

State (workspace)

`mydata`



user

Interfaces

Console (CLI)

`age_mean <-
mean(mydata$age)`

Button (GUI)

Clicks: Analyze > Descriptive Statistics >
Descriptives > add variable(s) > ok

same

Instruction (expression)

`age_mean <-
mean(mydata$age)`

R Engine / interpreter

1. lookup ``mydata`` in the workspace
2. lookup ``$age`` in ``mydata``
3. calculate mean of ``$age``
4. assign result to ``age_mean``

Output

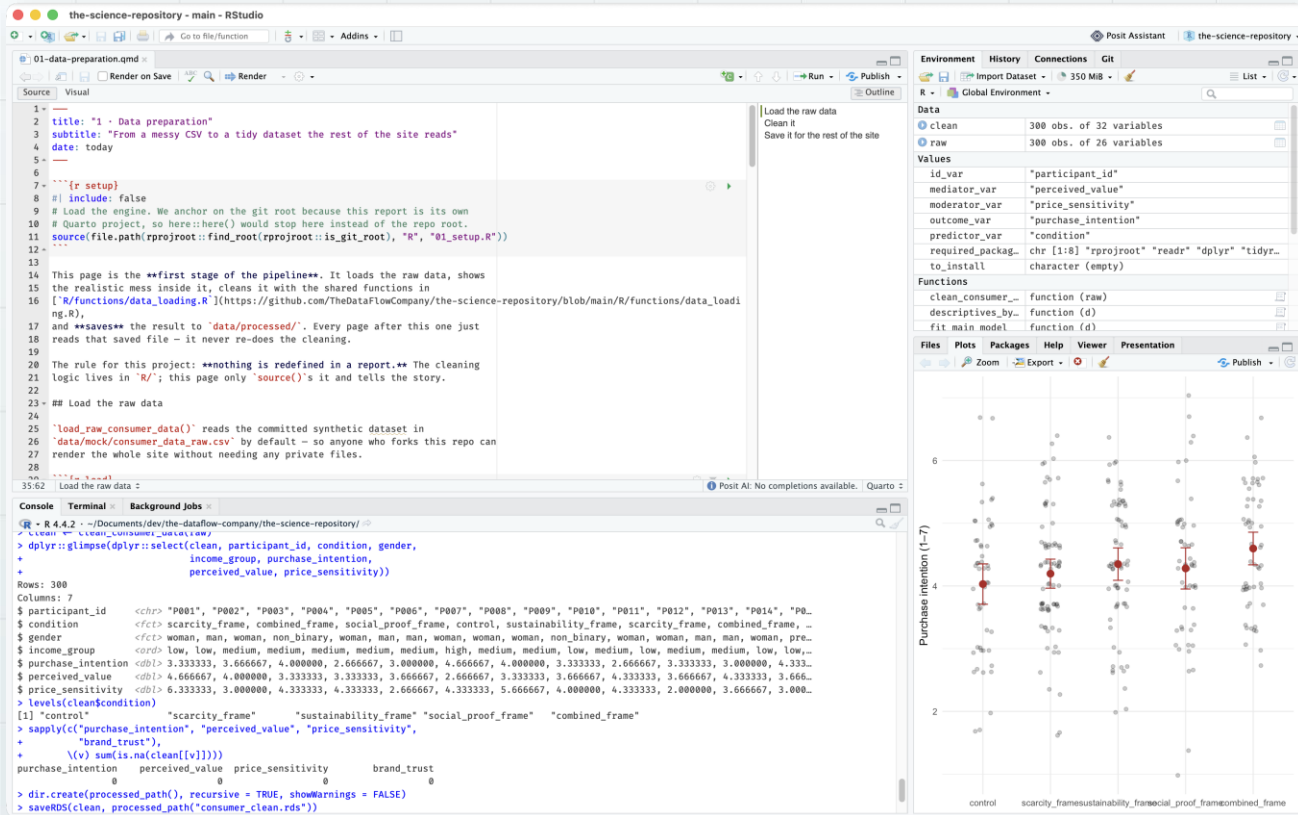
`#> 41`

State (workspace)

`mydata, age_mean`

```
where you write  
and save scripts
```

where R runs
R commands



the objects
in memory

```
manage files,  
see figures,  
read docs
```

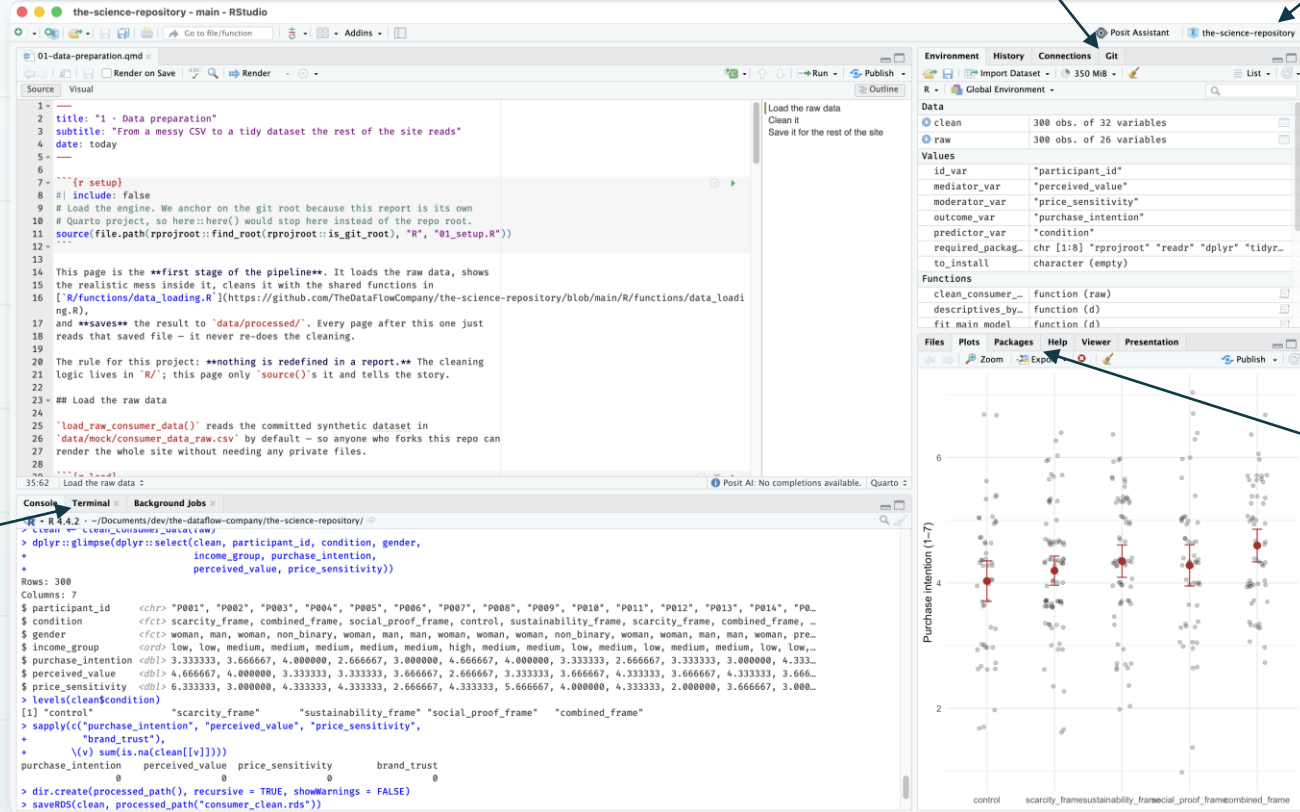
Git/GitHub

R project setup

your .Rproj settings

R Packages

Terminal
computer's
command line,
`quarto render`
command



Quarto = prose + code → polished output

A .qmd file mixes:

- text (Markdown)
- R code chunks
- the results those chunks produce (tables, figures, numbers)

Quarto runs the code and weaves results into the finished document.

Why Quarto (and not R Markdown)? Quarto is R Markdown's successor:

- one engine for R, Python, and Julia (not R-only)
- one source → HTML, PDF, Word, slides, whole websites
- actively developed, batteries included (citations, cross-references)

Practical

Best way of learning R is making mistakes. We are here to help you and teach you how to handle errors

- ❑ Option 1: The plots and analysis are rudimentary. Find a way to improve them. Then render them and look at your new webpage.
- ❑ Option 2: Collaborate with your neighbor. Learn how to use GitHub, branches, and merging to work together on two tasks: improving the moderation and mediation analyses at the same time.
- ❑ Option 3: Integrate your own data into the repository. Let's load, look at it, and run an analysis.

Day 3

The Report

Efficient Ins and Outs